

UNDERSTANDING IN-CONTEXT LEARNING FROM REPETITIONS

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ABSTRACT

This paper explores the elusive mechanism underpinning in-context learning in Large Language Models (LLMs). Our work provides a novel perspective by examining in-context learning via the lens of surface repetitions. We quantitatively investigate the role of surface features in text generation, and empirically establish the existence of *token co-occurrence reinforcement*, a principle that strengthens the relationship between two tokens based on their contextual co-occurrences. By investigating the dual impacts of these features, our research illuminates the internal workings of in-context learning and expounds on the reasons for its failures. This paper provides an essential contribution to the understanding of in-context learning and its potential limitations, providing a fresh perspective on this exciting capability.

1 INTRODUCTION

The impressive ability of Large Language Models (LLMs; Touvron et al. 2023a; Chowdhery et al. 2022; OpenAI 2023) to execute in-context learning (ICL) is a standout characteristic. This behavior mirrors human learning and reasoning from *analogy* (Winston, 1980), enabling LLMs to rapidly adapt to a range of downstream tasks. Without being explicitly pretrained to learn from demonstrations, LLMs can predict responses to unseen test queries from a few demonstrations and without any instruction given (Brown et al., 2020; Zhang et al., 2022; Chowdhery et al., 2022). An example of in-context learning can be found in Figure 1(a), where a pre-trained LLaMA model is given demonstrations for a binary classification task, and learns to make predictions correctly. Despite the success in applications, the working mechanism of in-context learning is still an open question.

Existing work has investigated input-label mapping (Min et al., 2022; Yoo et al., 2022; Wei et al., 2023) and demonstration construction (An et al., 2023; Lu et al., 2022; Liu et al., 2022) as underlying factors for ICL. However, little research has focused on the correlation between ICL and textual features. Intuitively, the behavior of ICL depends on the context and can be fragile to its variations. As Figure 1(b) shows, the same LLaMA model makes the incorrect prediction ‘True’ given the input “Circulation revenue has decreased by 5% in Finland.”, which is likely because of the repeated pattern “Answer:” -> “True” from the demonstrations. In the same perspective, the success case in Figure 1(a) can be attributed to learning desired patterns such as “Answer:” -> “True|False” in the demonstrations. Such patterns are apparently used as features in the autoregressive inference process by the model.

We take a feature-centric view to understand ICL, analyzing the key patterns in the input context that correlate with ICL behavior. The patterns we discussed above can be viewed as generalizations to repetition patterns (Holtzman et al., 2019; Fu et al., 2020) and self-reinforced patterns (Xu et al., 2022) which have been discussed in the literature. The ‘self-reinforcement effect’ describes a phenomenon where the model tends to perpetuate the generation of sentences that have frequently appeared in its context. These effects are regarded as harmful to text generation and previous work puts efforts



Figure 1: We showcase correct and incorrect predictions of in-context learning of LLaMA-65B. The shown task is to identify whether the given sentence presents a positive sentiment. We involve the token reinforced connections from demonstrations. In both cases, LLMs learn connections from the demonstrations and make decisions based on these connections. In the case of in-context learning, the model learns reliable connections and hopefully several of these connections result in the function of sentiment analysis. On the other hand, in repetitive demonstrations, the model gets stuck to spurious connections and misses the key information ‘*decreased*’, leading to a wrong prediction.

to mitigate it. However, they could give a unique view of the ICL behaviors from the angle of text generation

We quantitatively investigate in-context learning from the perspective of surface patterns, illustrating the inherent correlation among surface patterns, self-reinforcement, and ICL. First, we study the roles of self-reinforce patterns as surface features that guide text generation. We empirically establish the existence of the *token co-occurrence reinforcement*, where the connection between any two tokens gets reinforced with the number of contextual co-occurrences, a primary principle in learning surface-level patterns. We further delve into the reasons and inner-workings causing token reinforcement, showing it as an evitable result out of model’s efforts on maximizing likelihood on data.

Given the existence and reasons behind such patterns, we scrutinize the beneficial and detrimental effects of these surface patterns on in-context learning. On the one hand, experiments on MMLU and GSM8K show that the reinforcement helps constrain the output space and format outputs to follow demonstrations like outputting ‘Let’s think step by step.’. On the other hand, experiments with non-informative connections and reordered answers in MMLU demonstrate that intentionally constructed connections make LLMs lean towards specific answers, revealing the risk of unintended, spurious connections. This not only reveals the intrinsic workings of in-context learning to some extent, providing a perspective not analyzed in the literature but also explains the underlying reasons for the failure of in-context learning¹.

¹<https://github.com/ElliottYan/understand-icl-from-repetition>

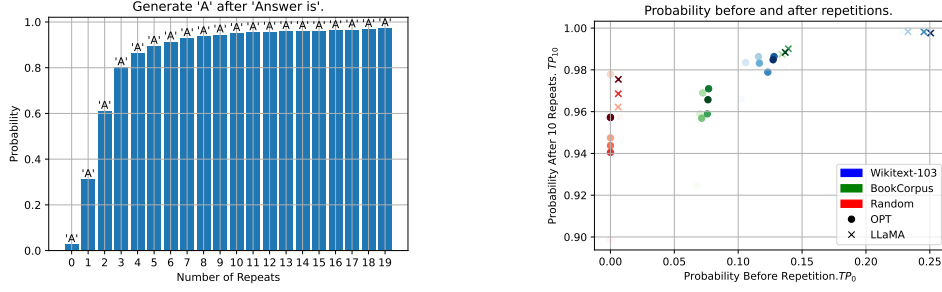


Figure 2: *Left: An example of the self-reinforcement effect.* We choose a normal sentence (‘Answer is **A**’), repeat it several times, and present the probability of the token ‘A’. The model used is LLaMA-7B. *Right: Sentence-level self-reinforcement over LLMs.* We plot all sizes of OPT and LLaMA with colors from light to dark. All sizes of LLaMA and OPT models demonstrate strong sentence-level self-reinforcement effects.

Our main contributions can be summarized as follows:

- We propose a novel perspective to understand ICL with repetitive text generations.
- We perform systematic analyses and empirically establish the existence of token reinforcement across various LLMs, alongside with the reason behind.
- We show that token reinforcement constrains output space and enables desired patterns for ICL, but is also responsible for spurious connections and possible failure of ICL.

2 ICL AND REPTITIONS

In ICL, given a desired task f , we feed an LLM with K demonstrations $\{(x_k, y_k), n \in [1, K]\}$, where $y_k = f(x_k)$. Here, y_k can be a label word or a free text phrase. Each demonstration $d^k = (\mathbf{F}_I, x_k, \mathbf{F}_O, y_k)$ can be divided into four parts. $\mathbf{F}_I$ and $\mathbf{F}_O$ denote formatting tokens for inputs and outputs, e.g., ‘Input:’ and ‘Answer:’. Note that x_k and y_k can consist of several tokens.

A pretrained language model $\mathcal{M}$ predicts the output y conditioned on the concatenation of both demonstrations and the test query x ,

$$\mathcal{P}_{\text{ICL}}(y|x, k) := \mathcal{M}(y | (\mathbf{F}_I, x_1, \mathbf{F}_O, y_1, \dots, \mathbf{F}_I, x_k, \mathbf{F}_O, y_k, \mathbf{F}_I, x, \mathbf{F}_O)). \quad (1)$$

To further understand the influence of surface repetitions from Figure 1(b), we show a case of sentence repetition and plots the probability of the sentence as the number of repetition in the previous context increases (Figure 2). When we manually repeat the sentence ‘Answer is A’, the probability of generating ‘A’ after ‘Answer is’ gets boosted from 0.03 to almost 1.0.

The right part of Figure 2 demonstrates our preliminary study with two families of large language models quantitatively, namely OPT (125M, 350M, 1.3B, 2.7B, 6.7B, 13B, 30B, Zhang et al. 2022) and LLaMA (7B, 13B, 30B, 65B, Touvron et al. 2023a). We plot the average sentence probability after 10 repeats for LLMs of varying sizes, arranged by light to dark colors. We use 1,000 sentences from each of the three different datasets – Wikitext-103 (Merity et al., 2016), BookCorpus (Zhu et al., 2015), and sequences of random words. More experimental details can be found in Appendix B. With 10 repeats, the probability of generating the sentence is significantly increased across all tested LLMs. Current LLMs amplify the occurrence of previously presented sentences, even *sequences of random tokens*.²

The above observations are related to the study of *self-reinforcement effect* (Xu et al., 2022) in literature. Formally, the conditional probability we model here is $\mathcal{P}_{\text{REP}}(w) :=$

²Xu et al. (2022) reported that sentences with a low initial probability — such as sentences composed of random tokens — have a smaller self-reinforcement effect. In our experiments, even sentences with random tokens (which initially have a near-zero probability) become reinforced to a probability nearing one. The difference may come from different model sizes (150M vs. maximum 65B) and pretrained corpus size (hundred millions of tokens vs. trillions of tokens).